

Dr. Tarun Kumar Kulshrestha

AI/ML Research Scientist | Deep Learning Engineer | Generative AI & Computer Vision

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PROFESSIONAL SUMMARY

Deep Learning and Generative AI researcher with 10+ years across applied research, engineering, and academic technology enablement. Ph.D. from IIT Roorkee and 5+ years at The Hong Kong Polytechnic University/UBDA. Experienced in converting research and proposal requirements into executable ML/DL workflows, guiding GPU/HPC experimentation, and publishing at CVPR, AAAI, WACV, IEEE TIP, and IEEE TMC. Expertise spans LLMs, RAG, GANs, computer vision, data-efficient learning, model optimization, and practical AI deployment.

CORE TECHNICAL SKILLS

AI / Deep Learning	Generative AI, LLMs/SLMs, RAG, LangChain, GANs, CNNs, RNNs, transfer learning, image-to-image translation, multimodal AI, few-shot/data-efficient learning, pruning, quantization, efficient inference
Frameworks & Programming	Python, PyTorch, TensorFlow, Keras, FastAPI, C, C++, Java
Platforms & Systems	Linux/Ubuntu, CUDA/GPU, HPC, Jupyter, Anaconda, Git/GitHub, VS Code, PyCharm, Cursor
Data & Applied AI	OCR, structured extraction, ETL, document intelligence, local models, Oracle, MySQL, Redis

PROFESSIONAL EXPERIENCE

Research Fellow

[University Research Facility in Big Data Analytics \(UBDA\), The Hong Kong Polytechnic University, Hong Kong](#) | Sep 2022 - Jun 2025

- Advised research teams on problem formulation, data preparation, model/framework selection, experimental design, GPU/HPC execution, debugging, performance analysis, and reproducible implementation.
- Designed, co-developed, and delivered hands-on training for PolyU researchers on LLMs, RAG/LangChain, GANs, image analytics, data analytics, CNN/RNN applications, medical imaging, and efficient image classification.
- Supported proposals and research plans with ML/DL methodology, data strategy, compute requirements, evaluation protocols, and implementation roadmaps.
- Conducted research on data-efficient and reconfigurable generative models, contributing to CVPR 2023, AAAI 2024, WACV 2025, IJCNN 2025, and ISDIA 2025.

Postdoctoral Fellow

[Department of Computing & UBDA, The Hong Kong Polytechnic University, Hong Kong](#) | Sep 2019 - Sep 2022

- Worked on RIF and ITF projects in food safety, big data, IoT, and non-targeted contaminant detection.
- Developed a smartphone-based, real-time, low-cost food-quality sensing prototype without external sensors; the demonstration received an A (Excellent) departmental evaluation.
- Researched GANs and unsupervised image-to-image translation, including model design, training, optimization, evaluation, and publication.

Senior Research Scholar / Research Scholar / Junior Research Fellow

[Indian Institute of Technology \(IIT\) Roorkee, India](#) | Jan 2015 - Aug 2019

- Developed SmartITS, a smartphone-sensor-cloud platform for seamless indoor-outdoor localization, crowd monitoring, trajectory tracking, and event detection.
- Built algorithms and proof-of-concept applications for spatio-temporal analytics, human-behaviour modelling, anomaly detection, and suspicious-activity identification.
- Published in IEEE Transactions on Mobile Computing, Journal of Network and Computer Applications, and IEEE conferences.

Earlier experience: Software Engineer, HCL Ltd. (Jul 2006 - Feb 2008); Assistant Professor roles at GLA University and Medi-Caps Institute.

SELECTED APPLIED AI & RESEARCH PROJECTS

- DocsMind - Document Intelligence: CPU-first pipeline for image/PDF → OCR → local language model → strict JSON → Excel; designed for privacy-aware document automation, field extraction, and modular document routing.
- MineSafe Training Intelligence: AI-assisted question generation, competency assessment, attendance analytics, worker/batch reports, and PDF report cards for industrial training programs.
- BusinessAgent OS: Product direction for a privacy-first multi-agent operating layer that converts messages and documents into leads, tasks, quotes, follow-ups, and business reports.
- Data-Efficient Generative Models: Architectural reconfiguration, regenerative pruning, layer-specific pruning, and dynamic adaptation for GANs under limited-data and limited-compute settings.
- Smartphone Food Quality Sensing: Real-time visual sensing and deep-learning pipeline for low-cost food freshness assessment.
- SmartITS Crowd Intelligence: Seamless indoor-outdoor localization, crowd sensing, trajectory tracking, event detection, and mobile-cloud analytics.

SELECTED POLYU / UBDA TRAINING & WORKSHOPS

- Foundations and Applications of Large Language Models (2025): LLM architectures, pre-training, Transformers, RAG, LangChain, and demonstrations.
- Image Analytics in Practice (2024): Transfer learning, GAN training, super-resolution, inpainting, restoration, and generative applications.
- Transforming Images: GAN-Based Image-to-Image Translation (2024) and Practical Data Analytics I/II.
- Efficient and Accurate Image Classification; RNNs for Text Generation; Deep Learning for Medical Image Analytics (2022).
- Scalable Parallel Computing: Machine Learning with GPU (2021).
- CNN/RNN Applications on UBDA; Data-Driven Research and Proposal Writing (2020).

SELECTED PUBLICATIONS

1. IJCNN 2025 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, Y. Zhao, "Dynamic Generative Adaptation for Data-Efficient GAN."
2. WACV 2025 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "Data-Efficient Alignment in Medical Imaging via Reconfigurable Generative Networks."
3. AAAI 2024 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "RG-GAN: Dynamic Regenerative Pruning for Data-Efficient Generative Adversarial Networks."
4. CVPR 2023 - D. Saxena, J. Cao, J. Xu, T. Kulshrestha, "Re-GAN: Data-Efficient GANs Training via Architectural Reconfiguration."
5. IEEE TIP 2022 - D. Saxena, T. Kulshrestha, J. Cao, S.-C. Cheung, "Multi-Constraint Adversarial Networks for Unsupervised Image-to-Image Translation."
6. IEEE TMC 2019 - T. Kulshrestha, D. Saxena, R. Niyogi, J. Cao, "Real-time Crowd Monitoring using Seamless Indoor-Outdoor Localization."
7. JNCA 2017 - T. Kulshrestha, D. Saxena, R. Niyogi, V. Raychoudhury, M. Misra, "SmartITS: Smartphone-based Identification and Tracking using Seamless Indoor-Outdoor Localization."

Full publication list: [Google Scholar](#)

EDUCATION, AWARDS & SERVICE

Ph.D., Computer Science & Engineering | Indian Institute of Technology (IIT) Roorkee, India | Jul 2015 - Aug 2019

- Best Paper Award - LS-GAN: Layer-Specific Pruning for Data-Efficient Generative Adversarial Networks, ISDIA 2025.
- Microsoft Research grant support, IIT Roorkee (2016); MHRD Scholarship for full-time Ph.D. studies.
- Reviewer: Nature, IEEE Access, Journal of Network and Computer Applications, and IEEE Sensors.

RESEARCH TRANSLATION & PROPOSAL SUPPORT

- Contributed ML/DL methodology, data/computing strategy, and implementation planning to the UBDA (Shenzhen) proposal.
- Contributed to the RGC Collaborative Research Fund equipment proposal "High Performance Deep Learning Clusters for Big Data Analytics" (2021/22).
- Through UBDA, supported researchers across PolyU departments in applying ML/DL to domain-specific projects, including method selection, experiment planning, implementation guidance, and proposal methodology.